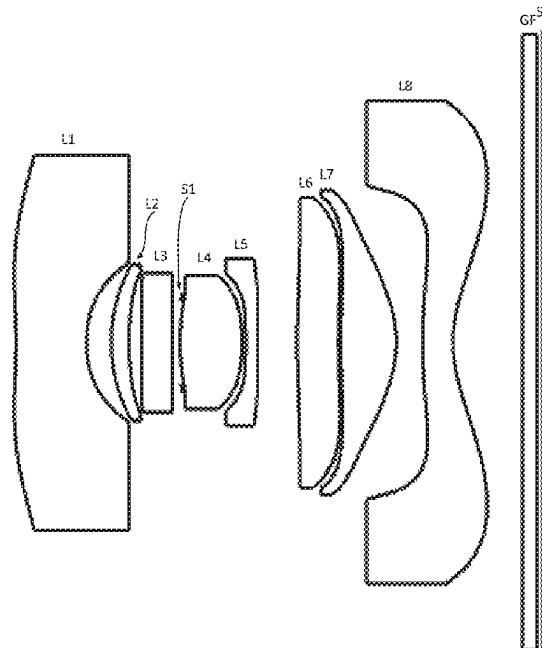


(45) **Date of Patent:** **Sep. 9, 2025**

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△

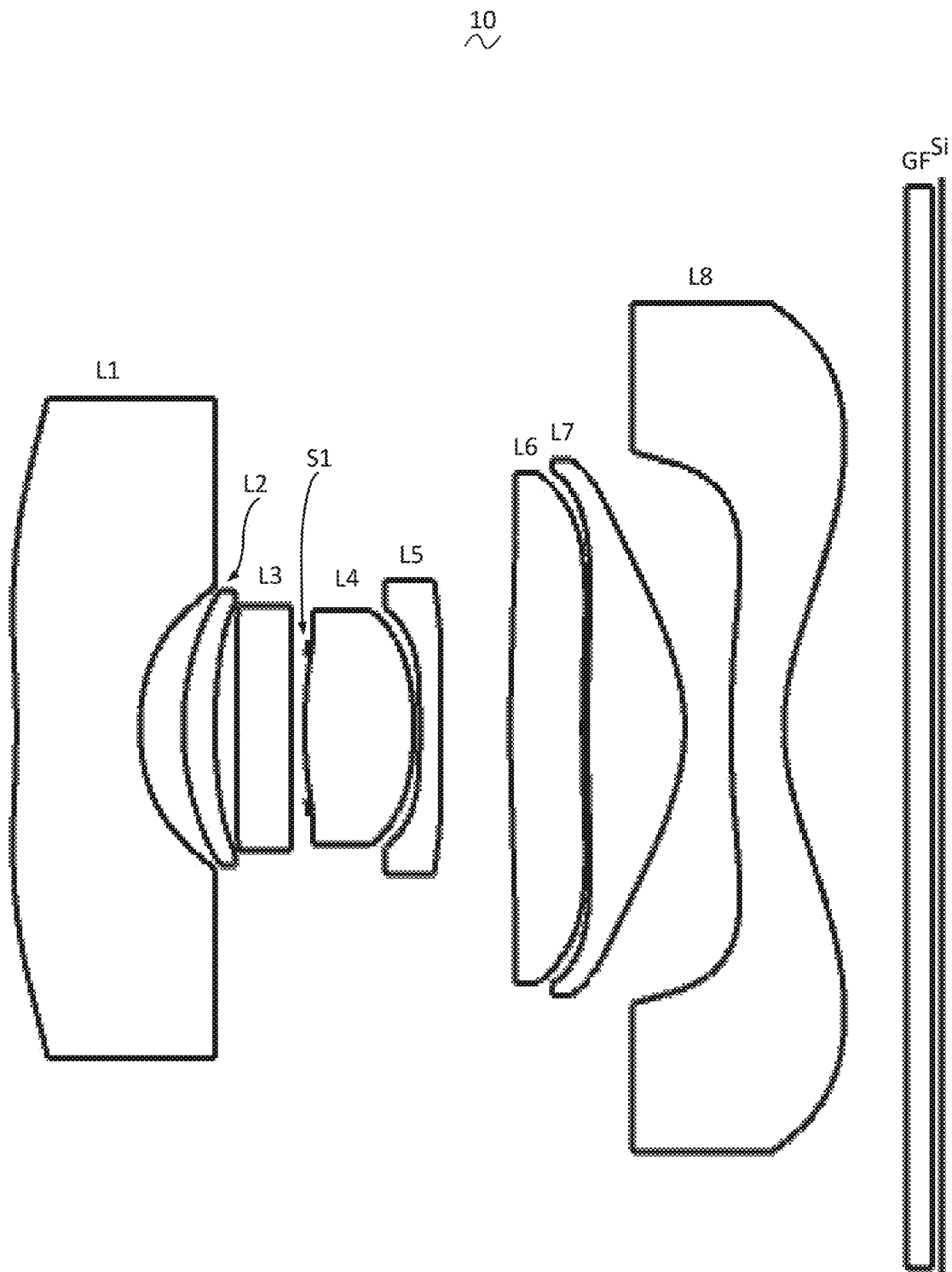


Fig. 1

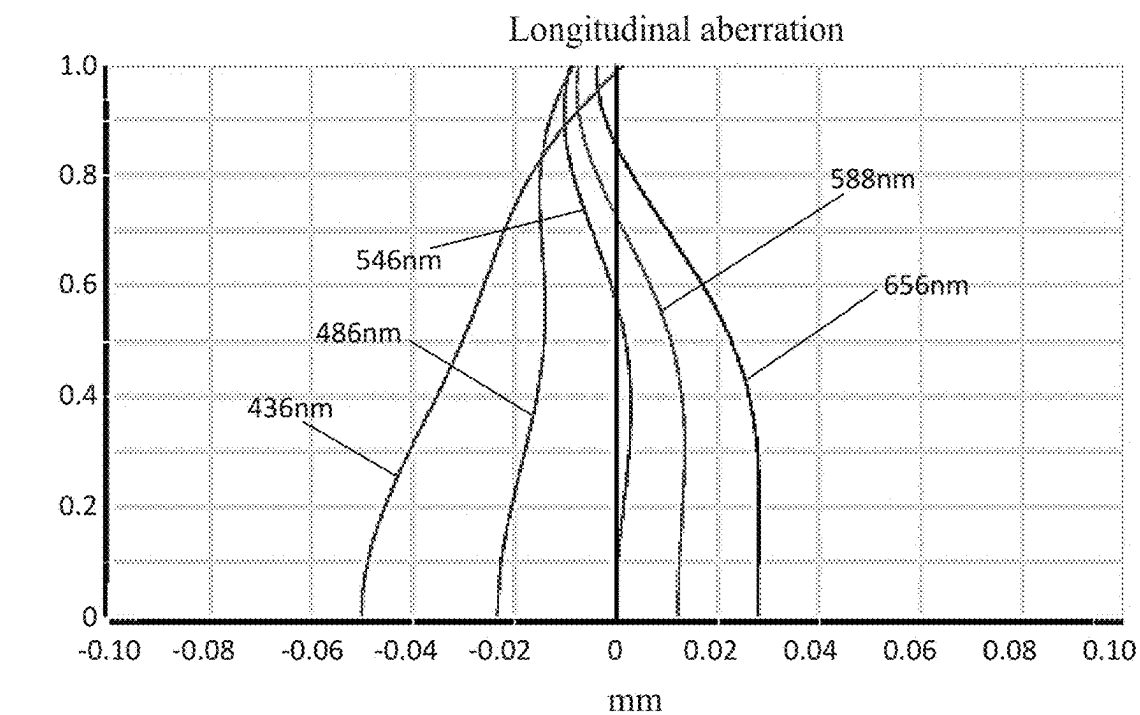


Fig. 2

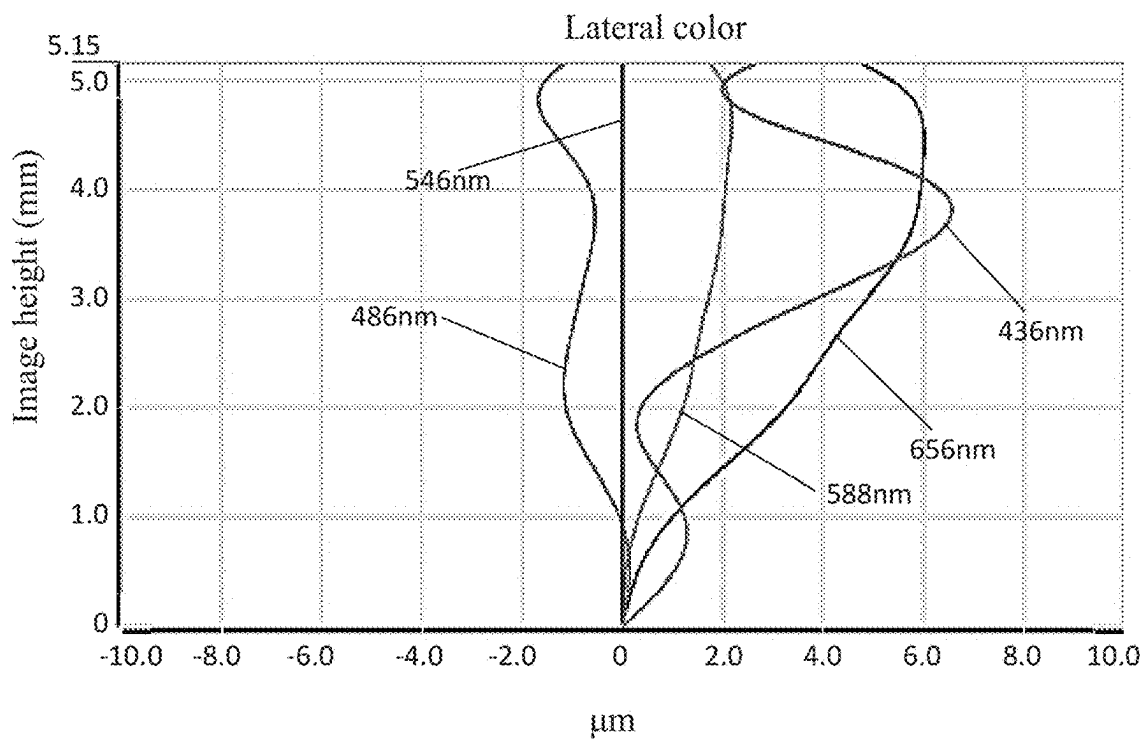


Fig. 3

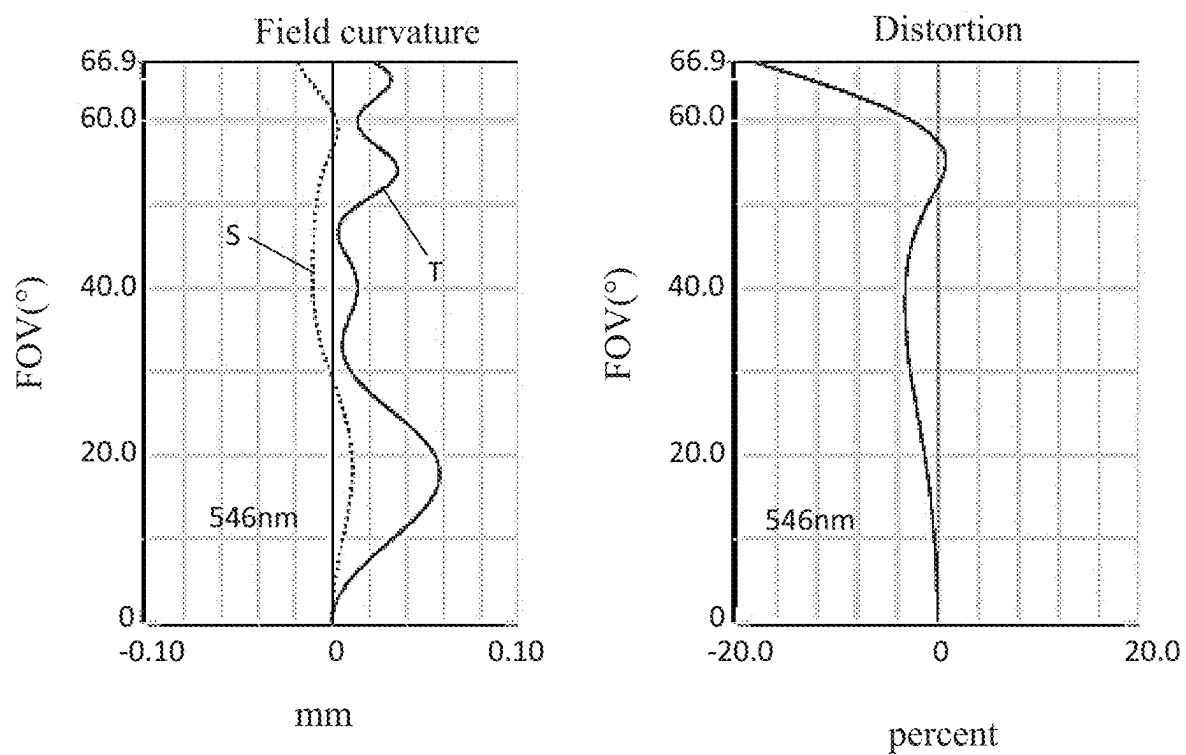


Fig. 4

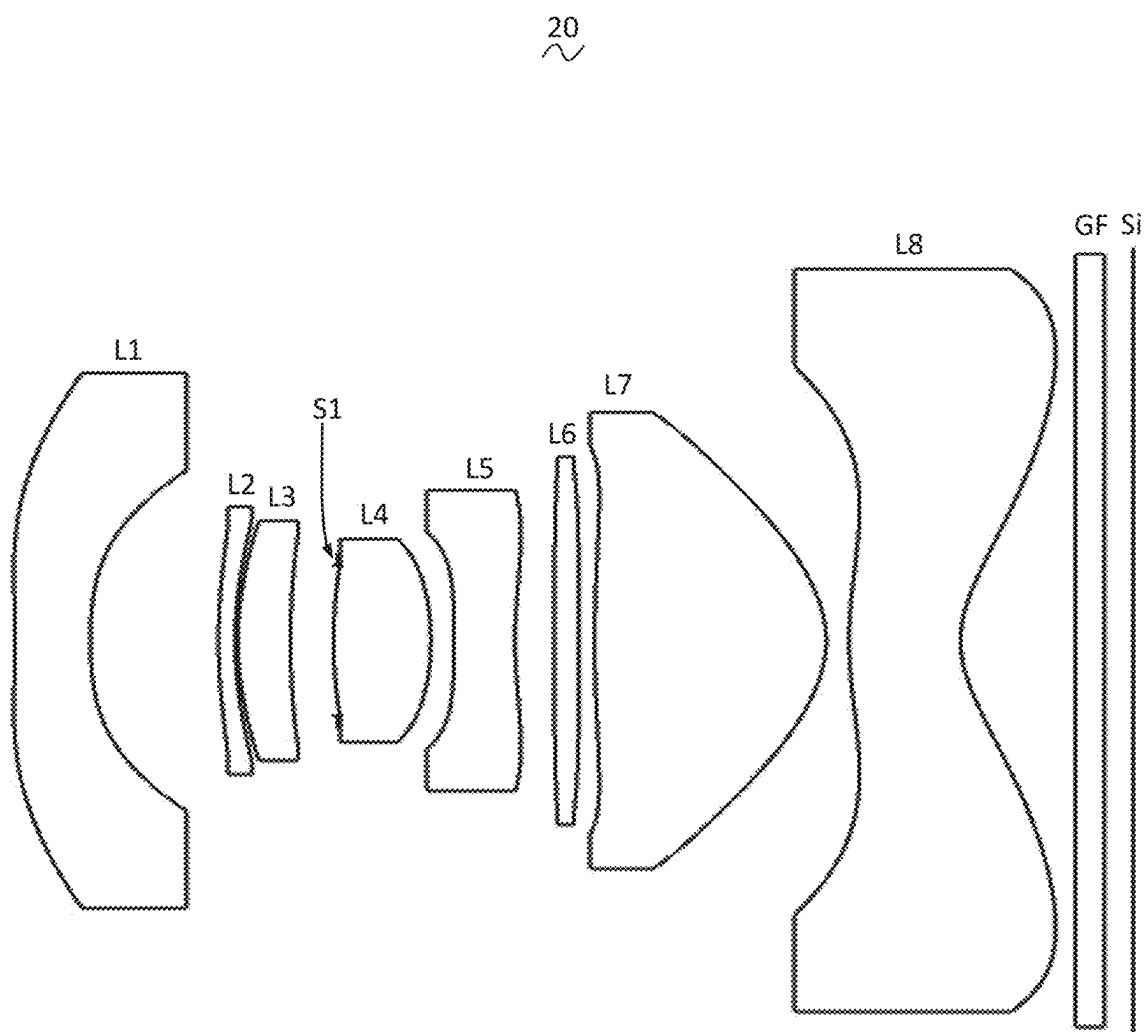


Fig. 5

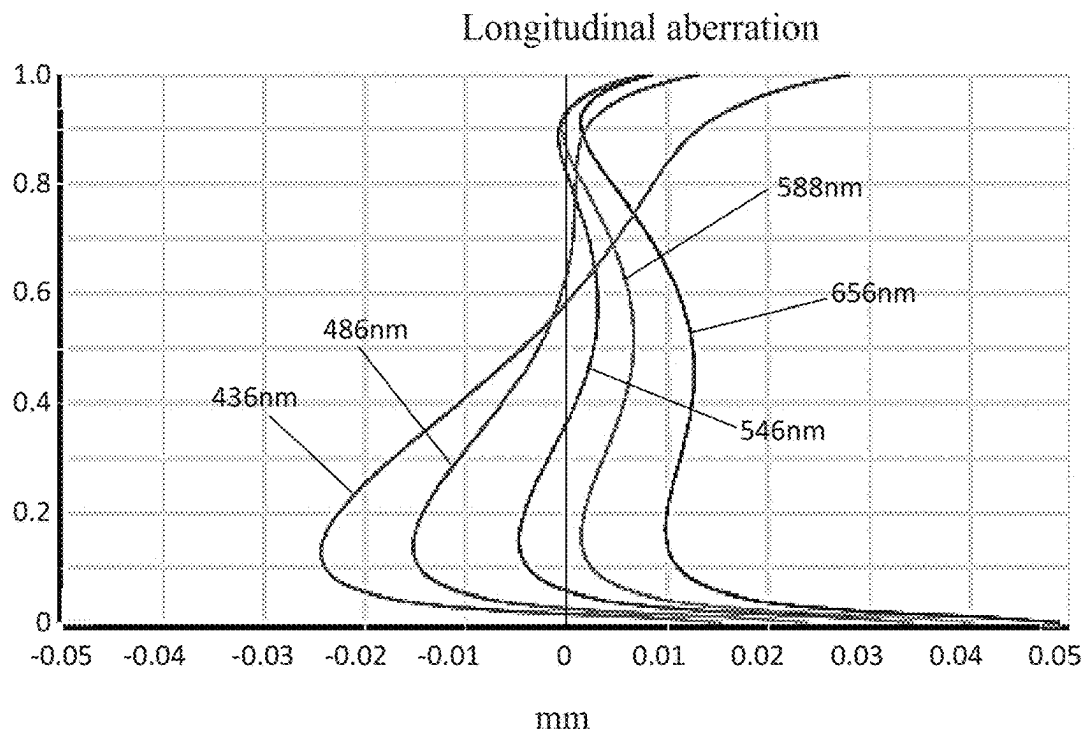


Fig. 6

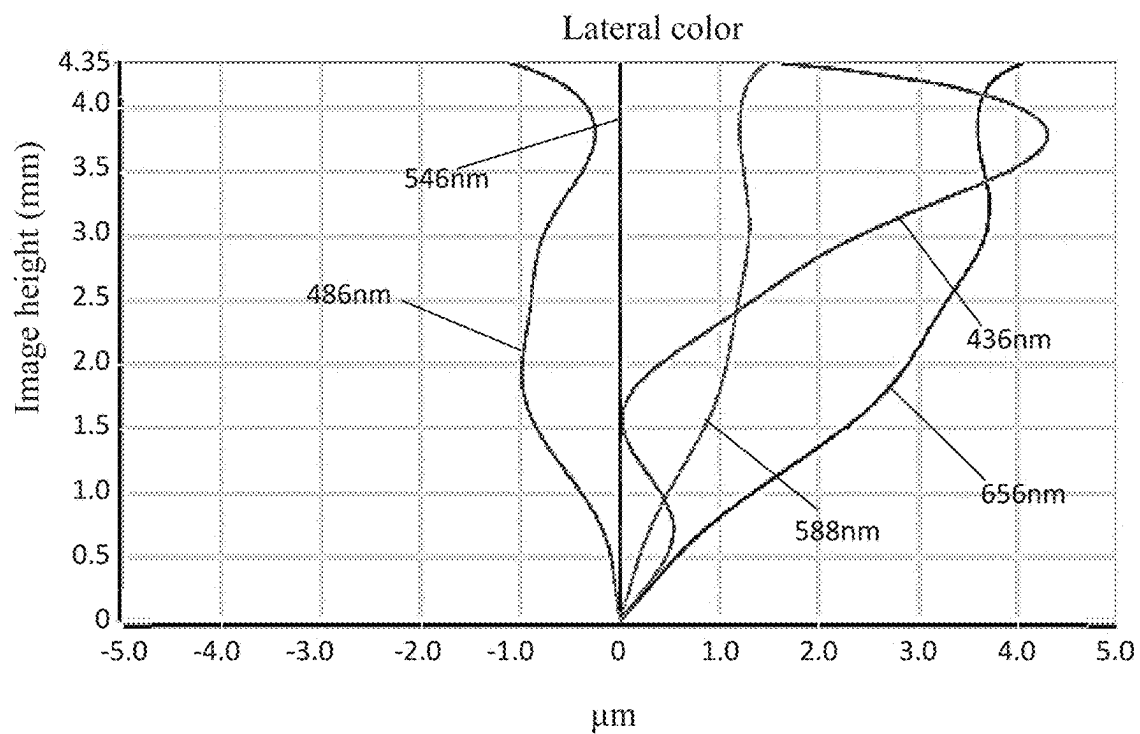


Fig. 7

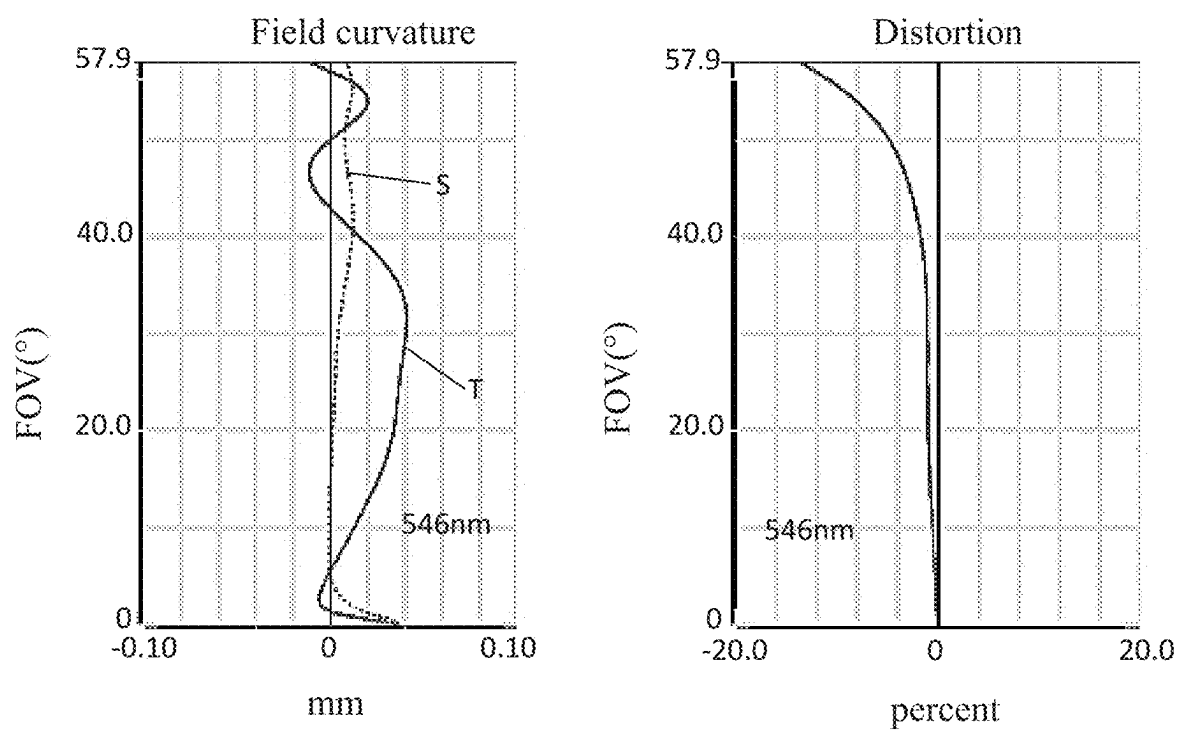


Fig. 8

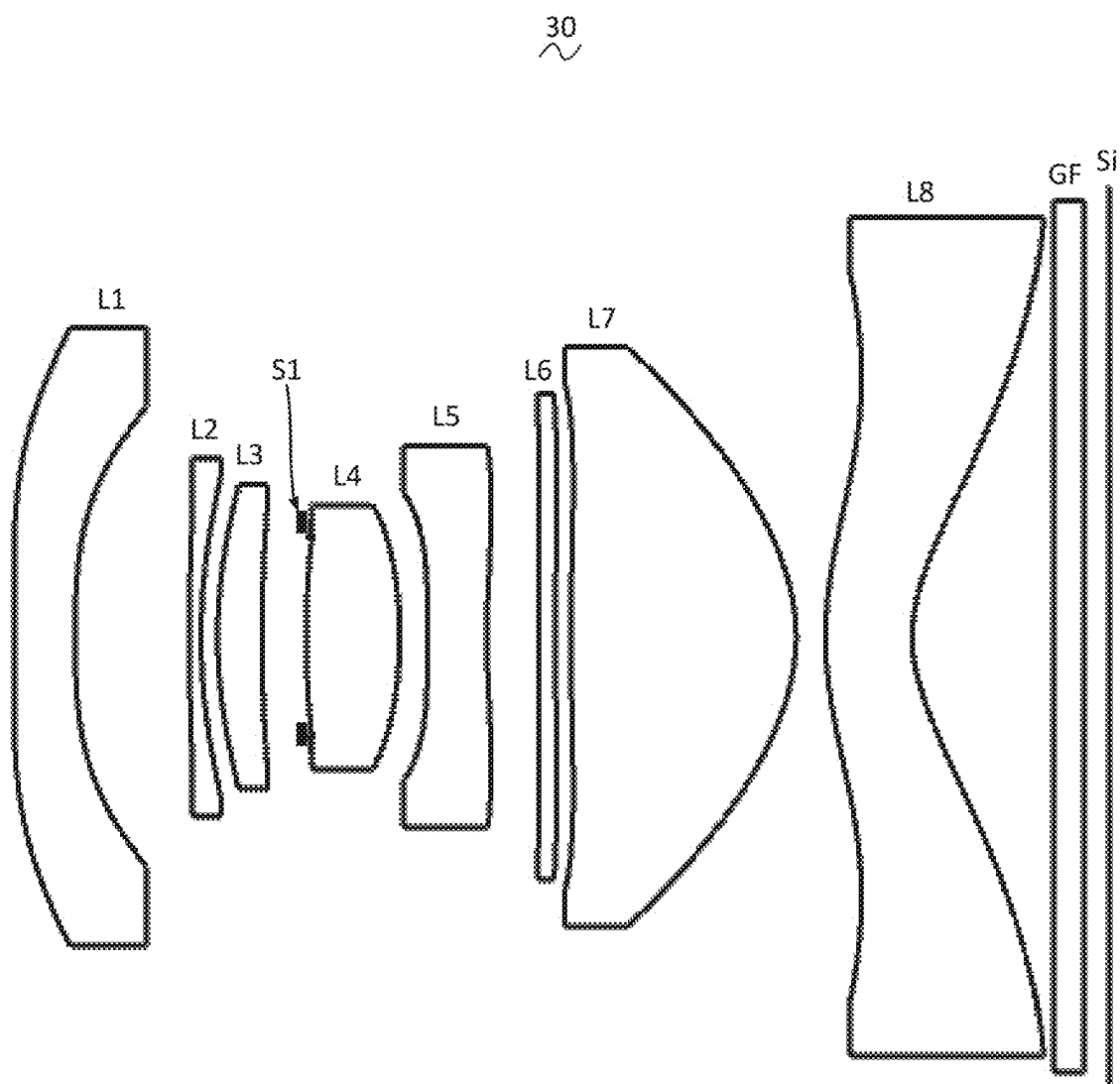


Fig. 9

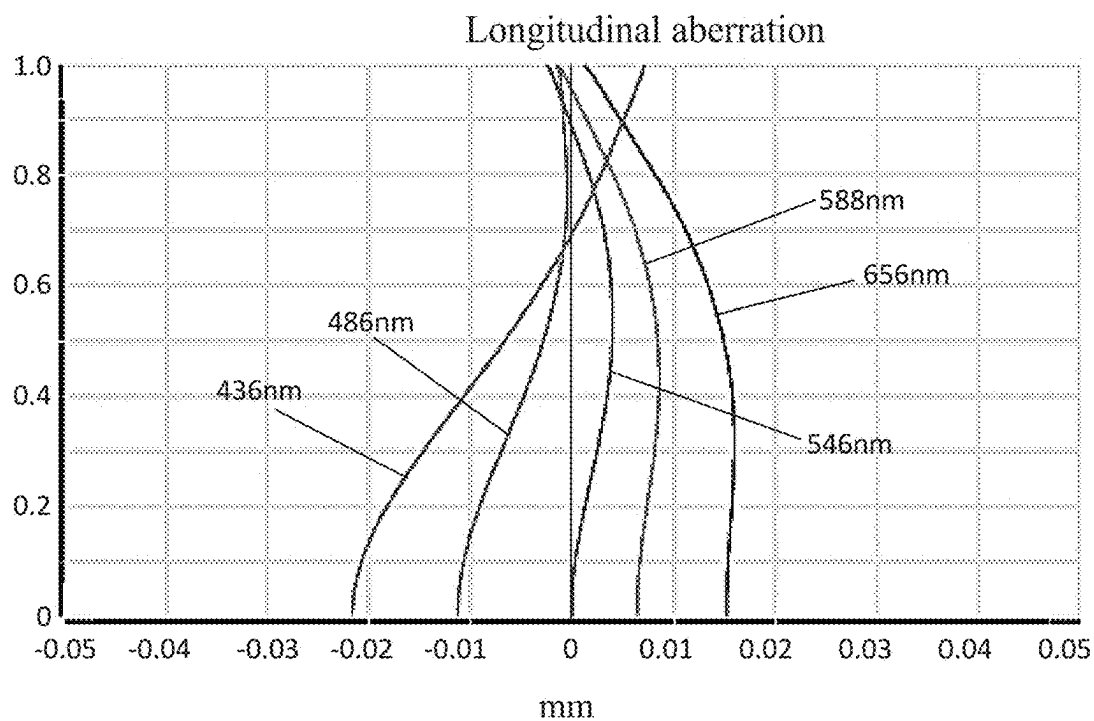


Fig. 10

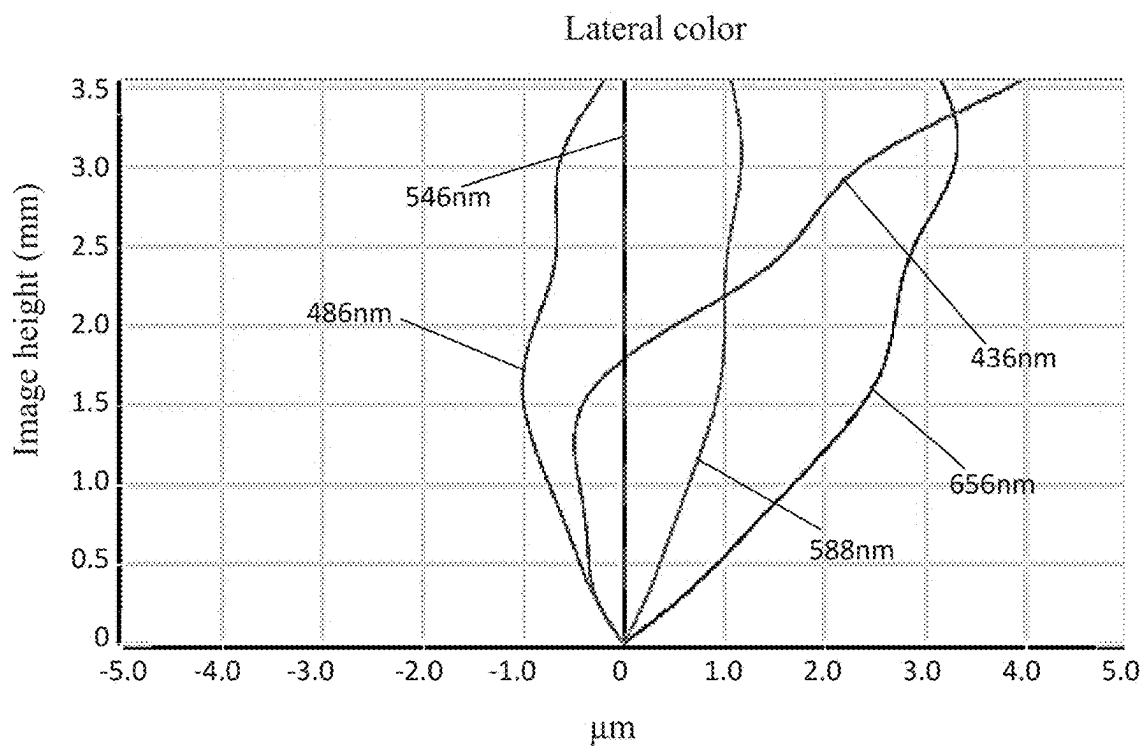


Fig. 11

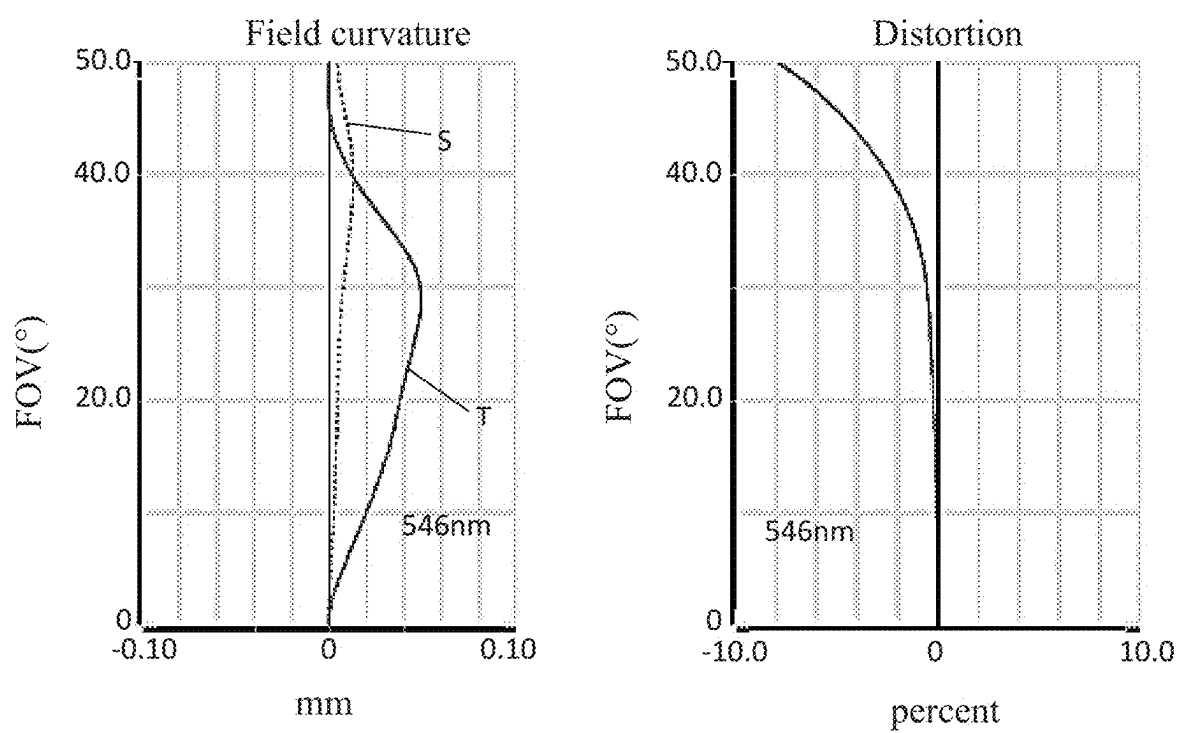


Fig. 12

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CAMERA OPTICAL LENS

FIELD OF THE PRESENT INVENTION

The present invention relates to the field of optical lens, and more particularly, to a camera optical lens suitable for handheld terminal devices, such as smart phones and digital cameras, monitors or PC lenses.

DESCRIPTION OF RELATED ART

In recent years, with the rise of various smart devices, the demand for miniaturized camera optics has been increasing, and the pixel size of photosensitive devices has shrunk, coupled with the development trend of electronic products with good functions, thin and portable appearance. Therefore, miniaturized imaging optical lenses with good image quality have become the mainstream in the current market. In order to obtain better imaging quality, a multi-piece lenses structure is often used. Moreover, with the development of technology and the increase of diversified needs of users, as the pixel area of the photosensitive device continues to shrink and the system's requirements for image quality continue to increase, the eight-piece lenses structure gradually appears in the lens design. There is an urgent need for a wide-angle imaging lens with excellent optical characteristics, small size, and fully corrected aberrations.

SUMMARY

In the present invention, a camera optical lens has a wide-angle effect.

According to one aspect of the present invention, a camera optical lens comprises, from an object side to an image side in sequence, a first lens having a negative refractive power, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, a seventh lens, and an eighth lens, wherein the camera optical lens satisfies the following conditions: $100.00^\circ \leq \text{FOV} \leq 135.00^\circ$, $-3.20 \leq f_1/f \leq -1.00$, and $2.30 \leq R_{15}/R_{16} \leq 5.00$. FOV denotes a field of view in a diagonal direction of the camera optical lens. f denotes a focal length of the camera optical lens, f_1 denotes a focal length of the first lens, R_{15} denotes a central curvature radius of an object side surface of the eighth lens, and R_{16} denotes a central curvature radius of an image side surface of the eighth lens.

As an improvement, the fifth lens has a negative refractive power. The camera optical lens satisfies the following condition: $-4.50 \leq f_5/f \leq -2.00$. f_5 denotes a focal length of the fifth lens.

As an improvement, the first lens has an object side surface being concave in a paraxial region and an image side surface being concave in the paraxial region. The camera optical lens further satisfies the following conditions: $0.15 \leq (R_1+R_2)/(R_1-R_2) \leq 0.88$ and $0.03 \leq d_1/\text{TTL} \leq 0.20$. R_1 denotes a central curvature radius of the object side surface of the first lens, R_2 denotes a central curvature radius of the image side surface of the first lens, d_1 denotes an on-axis thickness of the first lens, and TTL denotes a total optical length from the object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.

As an improvement, the camera optical lens further satisfies the following conditions: $0.23 \leq (R_1+R_2)/(R_1-R_2) \leq 0.71$ and $0.04 \leq d_1/\text{TTL} \leq 0.16$.

As an improvement, the second lens has an object side surface being convex in a paraxial region, an image side

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surface being concave in the paraxial region. The camera optical lens further satisfies the following conditions: $-17.11 \leq f_2/f \leq 4.12$, $-7.01 \leq (R_3+R_4)/(R_3-R_4) \leq 6.85$, and $0.01 \leq d_3/\text{TTL} \leq 0.05$. f_2 denotes a focal length of the second lens, R_3 denotes a central curvature radius of the object side surface of the second lens, R_4 denotes a central curvature radius of the image side surface of the second lens, d_3 denotes an on-axis thickness of the second lens, and TTL denotes a total optical length from an object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.

As an improvement, the camera optical lens further satisfies the following conditions: $-10.69 \leq f_2/f \leq 3.30$, $-4.38 \leq (R_3+R_4)/(R_3-R_4) \leq 5.48$, and $0.01 \leq d_3/\text{TTL} \leq 0.04$.

As an improvement, the third lens has an object side surface being convex in a paraxial region, an image side surface being concave in the paraxial region, and the third lens has a positive refractive power. The camera optical lens further satisfies the following conditions: $1.21 \leq f_3/f \leq 408.22$, $-20.33 \leq (R_5+R_6)/(R_5-R_6) \leq -1.27$, and $0.02 \leq d_5/\text{TTL} \leq 0.09$. f_3 denotes a focal length of the third lens, R_5 denotes a central curvature radius of the object side surface of the third lens, R_6 denotes a central curvature radius of the image side surface of the third lens, d_5 denotes an on-axis thickness of the third lens, and TTL denotes a total optical length from an object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.

As an improvement, the camera optical lens further satisfies the following conditions: $1.93 \leq f_3/f \leq 326.58$, $-12.70 \leq (R_5+R_6)/(R_5-R_6) \leq -1.58$, and $0.03 \leq d_5/\text{TTL} \leq 0.07$.

As an improvement, the fourth lens has an object side surface being convex in a paraxial region, an image side surface being convex in the paraxial region, and the fourth lens has a positive refractive power. The camera optical lens further satisfies the following conditions: $0.59 \leq f_4/f \leq 2.14$, $0.07 \leq (R_7+R_8)/(R_7-R_8) \leq 0.46$, and $0.04 \leq d_7/\text{TTL} \leq 0.18$. f_4 denotes a focal length of the fourth lens, R_7 denotes a central curvature radius of the object side surface of the fourth lens, R_8 denotes a central curvature radius of the image side surface of the fourth lens, d_7 denotes an on-axis thickness of the fourth lens, and TTL denotes a total optical length from an object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.

As an improvement, the camera optical lens further satisfies the following conditions: $0.95 \leq f_4/f \leq 1.71$, $0.11 \leq (R_7+R_8)/(R_7-R_8) \leq 0.37$, and $0.07 \leq d_7/\text{TTL} \leq 0.14$.

As an improvement, the fifth lens has an object side surface being concave in a paraxial region and an image side surface being concave in the paraxial region. The camera optical lens further satisfies the following conditions: $0.37 \leq (R_9+R_{10})/(R_9-R_{10}) \leq 1.42$ and $0.01 \leq d_9/\text{TTL} \leq 0.08$. R_9 denotes a central curvature radius of the object side surface of the fifth lens, R_{10} denotes a central curvature radius of the image side surface of the fifth lens, d_9 denotes an on-axis thickness of the fifth lens, and TTL denotes a total optical length from an object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.

As an improvement, the camera optical lens further satisfies the following conditions: $0.60 \leq (R_9+R_{10})/(R_9-R_{10}) \leq 1.14$ and $0.02 \leq d_9/\text{TTL} \leq 0.07$.

As an improvement, the camera optical lens further satisfies the following conditions: $-645.45 \leq f_6/f \leq 138.30$, $0.35 \leq (R_{11}+R_{12})/(R_{11}-R_{12}) \leq 64.63$, and $0.01 \leq d_{11}/\text{TTL} \leq 0.12$. f_6 denotes a focal length of the sixth lens, R_{11} denotes a central

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curvature radius of an object side surface of the sixth lens, R12 denotes a central curvature radius of an image side surface of the sixth lens, d11 denotes an on-axis thickness of the sixth lens, and TTL denotes a total optical length from an object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.

As an improvement, the camera optical lens further satisfies the following conditions: $-403.41 \leq f_6/f \leq 110.64$, $0.57 \leq (R11+R12)/(R11-R12) \leq 51.70$, and $0.01 \leq d11/TTL \leq 0.09$.

As an improvement, the seventh lens has an object side surface being convex in a paraxial region and an image side surface being convex in a paraxial region, and the seventh lens has a positive refractive power. The camera optical lens further satisfies the following conditions: $0.40 \leq f_7/f \leq 1.59$, $0.38 \leq (R13+R14)/(R13-R14) \leq 1.29$, and $0.05 \leq d13/TTL \leq 0.31$. f7 denotes a focal length of the seventh lens, R13 denotes a central curvature radius of the object side surface of the seventh lens, R14 denotes a central curvature radius of the image side surface of the seventh lens, d13 denotes an on-axis thickness of the seventh lens, and TTL denotes a total optical length from an object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.

As an improvement, the camera optical lens further satisfies the following conditions: $0.64 \leq f_7/f \leq 1.27$, $0.60 \leq (R13+R14)/(R13-R14) \leq 1.03$, and $0.08 \leq d13/TTL \leq 0.25$.

As an improvement, the object side surface of the eighth lens is convex in a paraxial region, the image side surface of the eighth lens is concave in the paraxial region, and the eighth lens has a negative refractive power. The camera optical lens further satisfies the following conditions: $-2.96 \leq f_8/f \leq -0.69$, $0.75 \leq (R15+R16)/(R15-R16) \leq 3.80$, and $0.03 \leq d15/TTL \leq 0.15$. f8 denotes a focal length of the eighth lens, d15 denotes an on-axis thickness of the eighth lens, and TTL denotes a total optical length from an object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.

As an improvement, the camera optical lens further satisfies the following conditions: $-1.85 \leq f_8/f \leq -0.86$, $1.20 \leq (R15+R16)/(R15-R16) \leq 3.04$, and $0.05 \leq d15/TTL \leq 0.12$.

As an improvement, the camera optical lens further satisfies the following condition: $TTL/IH \leq 2.91$. IH denotes an image height of the camera optical lens, and TTL denotes a total optical length from an object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis.

As an improvement, an FNO of the camera optical lens is less than or equal to 2.68. FNO denotes a ratio of an effective focal length of the camera optical lens to an entrance pupil diameter.

BRIEF DESCRIPTION OF THE DRAWINGS

In order to explain the technical solutions in the embodiments of the present invention more clearly, the following will briefly introduce the drawings that need to be used in the description of the embodiments. Obviously, the drawings in the following description are only some embodiments of the present invention. For those of ordinary skill in the art, without creative work, other drawings can be obtained based on these drawings, among which:

FIG. 1 is a schematic diagram of a structure of a camera optical lens in accordance with Embodiment 1 of the present invention;

FIG. 2 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 1;

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FIG. 3 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 1;

FIG. 4 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 1;

FIG. 5 is a schematic diagram of a structure of a camera optical lens in accordance with Embodiment 2 of the present invention;

FIG. 6 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 5;

FIG. 7 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 5;

FIG. 8 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 5;

FIG. 9 is a schematic diagram of a structure of a camera optical lens in accordance with Embodiment 3 of the present invention;

FIG. 10 is a schematic diagram of a longitudinal aberration of the camera optical lens shown in FIG. 9;

FIG. 11 is a schematic diagram of a lateral color of the camera optical lens shown in FIG. 9; and

FIG. 12 is a schematic diagram of a field curvature and a distortion of the camera optical lens shown in FIG. 9.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

In order to make the objects, technical solutions, and advantages of the present invention more apparent, the embodiments of the present invention will be described in detail below. However, it will be apparent to the one skilled in the art that, in the various embodiments of the present invention, a number of technical details are presented in order to provide the reader with a better understanding of the invention. However, the technical solutions claimed in the present invention can be implemented without these technical details and various changes and modifications based on the following embodiments.

Embodiment 1

As referring to the accompanying drawings, the present invention provides a camera optical lens 10. FIG. 1 shows the camera optical lens 10 according to embodiment 1 of the present invention. The camera optical lens 10 comprises eight-piece lenses. Specifically, from an object side to an image side, the camera optical lens 10 comprises in sequence: a first lens L1, a second lens L2, a third lens L3, an aperture S1, a fourth lens L4, a fifth lens L5, a sixth lens L6, a seventh lens L7 and an eighth lens L8. Optical elements like optical filter GF can be arranged between the eighth lens L8 and an image surface S1.

The first lens L1 is made of plastic material, the second lens L2 is made of plastic material, the third lens L3 is made of plastic material, the fourth lens L4 is made of plastic material, the fifth lens L5 is made of plastic material, the sixth lens L6 is made of plastic material, the seventh lens L7 is made of plastic material, and the eighth lens L8 is made of plastic material.

In other optional embodiments, each lens may also be made of other materials, such as glasses.

In the present embodiment, a field of view of the camera optical lens 10 in a diagonal direction is defined as FOV, and the camera optical lens 10 satisfies the following condition: $100.00^\circ \leq \text{FOV} \leq 135.00^\circ$. The range of the FOV is specified. When the camera optical lens 10 satisfies the above condition, the camera optical lens 10 has a wide-angle effect.

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A focal length of the camera optical lens **10** is defined as f , and a focal length of the first lens **L1** is defined as f_1 . The camera optical lens **10** meets the following condition: $-3.20 \leq f_1/f \leq -1.00$. A ratio of the focal length f_1 of the first lens **L1** to a focal length f of the camera optical lens **10** is specified. When the above condition is satisfied, the performance of the camera optical lens **10** would be improved.

A central curvature radius of an object side surface of the eighth lens **L8** is defined as R_{15} , and a central curvature radius of an image side surface of the eighth lens **L8** is defined as R_{16} . The camera optical lens **10** meets the following condition: $2.30 \leq R_{15}/R_{16} \leq 5.00$. The condition specifies a shape of the eighth lens **L8**. When the condition is satisfied, a light deflection can be alleviated while the light entry into the lens, and an aberration is effectively reduced.

The focal length of the camera optical lens **10** is defined as f , and a focal length of the fifth lens **L5** is defined as f_5 . The camera optical lens **10** meets the following condition: $-4.50 \leq f_5/f \leq -2.00$. A ratio of the focal length f_5 of the fifth lens **L5** to the focal length f of the camera optical lens is specified in the condition. The camera optical lens **10** has a higher imaging quality and a lower sensitivity as the focal length reasonably distributed. Preferably, the camera optical lens **10** further meets the following condition: $-4.38 \leq f_5/f \leq -2.19$.

In the present embodiment, an object side surface of the first lens **L1** is concave in a paraxial region, an image side surface of the first lens **L1** is concave in the paraxial region, and the first lens **L1** has a negative refractive power. In other optional embodiments, the object side surface and the image side surface of the first lens **L1** can also be arranged as other concave side surface or convex side surface, such as, convex object side surface and convex image side surface and so on.

A central curvature radius of the object side surface of the first lens **L1** is defined as R_1 , and a central curvature radius of the image side surface of the first lens **L1** is defined as R_2 . The camera optical lens **10** further satisfies the following condition: $0.15 \leq (R_1+R_2)/(R_1-R_2) \leq 0.88$. This condition reasonably controls a shape of the first lens **L1**, so that the first lens **L1** can effectively correct a spherical aberration of the camera optical lens **10**. Preferably, the following condition shall be satisfied, $0.23 \leq (R_1+R_2)/(R_1-R_2) \leq 0.71$.

An on-axis thickness of the first lens **L1** is defined as d_1 . A total optical length from the object side surface of the first lens **L1** to the image surface **S1** of the camera optical lens **10** along an optical axis is defined as TTL. The camera optical lens **10** further satisfies the following condition: $0.03 \leq d_1/\text{TTL} \leq 0.20$. When the value is within this range, it is beneficial for producing an ultra-thin effect. Preferably, the following condition shall be satisfied, $0.04 \leq d_1/\text{TTL} \leq 0.16$.

In the present embodiment, an object side surface of the second lens **L2** is convex in the paraxial region, the image side surface of the second lens **L2** is concave in the paraxial region, and the second lens **L2** has a positive refractive power. In other optional embodiments, the object side surface and the image side surface of the second lens **L2** can also be arranged as other concave side surface or convex side surface, such as, concave object side surface and convex image side surface and so on.

The focal length of the camera optical lens **10** is defined as f , and a focal length of the second lens **L2** is defined as f_2 . The camera optical lens **10** further satisfies the following condition: $-17.11 \leq f_2/f \leq 4.12$. It is beneficial for correcting the aberration of the camera optical lens **10** by controlling the positive refractive power of the second lens **L2** being

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within reasonable range. Preferably, the following condition shall be satisfied, $-10.69 \leq f_2/f \leq 3.30$.

A central curvature radius of the object side surface of the second lens **L2** is defined as R_3 , and a central curvature radius of the image side surface of the second lens **L2** is defined as R_4 . The camera optical lens **10** further satisfies the following condition: $-7.01 \leq (R_3+R_4)/(R_3-R_4) \leq 6.85$, which specifies a shape of the second lens **L2**. When the condition is satisfied, as the camera optical lens **10** develops toward ultra-thin and wide-angle lenses, it is beneficial for correcting an on-axis chromatic aberration. Preferably, the following condition shall be satisfied, $-4.38 \leq (R_3+R_4)/(R_3-R_4) \leq 5.48$.

An on-axis thickness of the second lens **L2** is defined as d_3 . The total optical length from the object side surface of the first lens **L1** to the image surface **S1** of the camera optical lens **10** along the optical axis is defined as TTL. The camera optical lens **10** further satisfies the following condition: $0.01 \leq d_3/\text{TTL} \leq 0.05$. When the value is within this range, it is beneficial for producing the ultra-thin effect. Preferably, the following condition shall be satisfied, $0.01 \leq d_3/\text{TTL} \leq 0.04$.

In the present embodiment, an object side surface of the third lens **L3** is convex in the paraxial region, an image side surface of the third lens **L3** is concave in the paraxial region, and the third lens **L3** has a positive refractive power. In other optional embodiments, the object side surface and the image side surface of the third lens **L3** can also be arranged as other concave side surface or convex side surface, such as, concave object side surface and convex image side surface and so on.

The focal length of the camera optical lens **10** is defined as f , and a focal length of the third lens **L3** is defined as f_3 . The camera optical lens **10** further satisfies the following condition: $1.21 \leq f_3/f \leq 408.22$. By a reasonable distribution of the refractive power, which makes the camera optical lens **10** has the higher imaging quality and the lower sensitivity. Preferably, the following condition shall be satisfied, $1.93 \leq f_3/f \leq 326.58$.

A central curvature radius of the object side surface of the third lens **L3** is defined as R_5 , and a central curvature radius of the image side surface of the third lens **L3** is defined as R_6 . The camera optical lens **10** further satisfies the following condition: $-20.33 \leq (R_5+R_6)/(R_5-R_6) \leq -1.27$, which specifies a shape of the third lens **L3**. It is beneficial for molding the third lens **L3**. When the condition is satisfied, a light deflection can be alleviated while the light passing through the lens, and the aberration can be reduced effectively. Preferably, the following condition shall be satisfied, $-12.70 \leq (R_5+R_6)/(R_5-R_6) \leq -1.58$.

An on-axis thickness of the third lens **L3** is defined as d_5 . The total optical length from the object side surface of the first lens **L1** to the image surface **S1** of the camera optical lens **10** along the optical axis is defined as TTL. The camera optical lens **10** further satisfies the following condition: $0.02 \leq d_5/\text{TTL} \leq 0.09$, which benefits for realizing the ultra-thin effect. Preferably, the following condition shall be satisfied, $0.03 \leq d_5/\text{TTL} \leq 0.07$.

In the present embodiment, the object side surface of the fourth lens **L4** is convex in the paraxial region, an image side surface of the fourth lens **L4** is convex in the paraxial region, and the fourth lens **L4** has a positive refractive power. In other optional embodiments, the object side surface and the image side surface of the fourth lens **L4** can also be arranged as other convex side surface or concave side surface, such as, concave object side surface and concave image side surface and so on.

The focal length of the camera optical lens **10** is defined as f , and a focal length of the fourth lens **L4** is defined as f_4 . The camera optical lens **10** further satisfies the following condition: $0.59 \leq f_4/f \leq 2.14$. It is beneficial for realizing the excellent imaging quality and the lower sensitivity by controlling the refractive power being within reasonable range. Preferably, the following condition shall be satisfied, $0.95 \leq f_4/f \leq 1.71$.

A curvature radius of the object side surface of the fourth lens **L4** is defined as R_7 , and a central curvature radius of the image side surface of the fourth lens **L4** is defined as R_8 . The camera optical lens further satisfies the following condition: $0.07 \leq (R_7+R_8)/(R_7-R_8) \leq 0.46$, which specifies a shape of the fourth lens **L4**. When the condition is satisfied, as the development of the ultra-thin and wide-angle lenses, it is beneficial for solving the problems, such as correcting an off-axis aberration. Preferably, the following condition shall be satisfied, $0.11 \leq (R_7+R_8)/(R_7-R_8) \leq 0.37$.

An on-axis thickness of the fourth lens **L4** is defined as d_7 . The total optical length from the object side surface of the first lens **L1** to the image surface **S1** of the camera optical lens **10** along the optical axis is defined as TTL . The camera optical lens **10** further satisfies the following condition: $0.04 \leq d_7/TTL \leq 0.18$, which is beneficial for realizing the ultra-thin effect. Preferably, the following condition shall be satisfied, $0.07 \leq d_7/TTL \leq 0.14$.

In the present embodiment, an object side surface of the fifth lens **L5** is concave in the paraxial region, an image side surface of the fifth lens **L5** is concave in the paraxial region, and the fifth lens **L5** has a negative refractive power. In other optional embodiments, the object side surface and the image side surface of the fifth lens **L5** can also be arranged as other convex side surface or concave side surface, such as, convex object side surface and convex image side surface and so on. The fifth lens **L5** may have a positive refractive power.

A central curvature radius of the object side surface of the fifth lens **L5** is defined as R_9 , and a central curvature radius of the image side surface of the fifth lens **L5** is defined as R_{10} . The camera optical lens further satisfies the following condition: $0.37 \leq (R_9+R_{10})/(R_9-R_{10}) \leq 1.42$, which specifies a shape of the fifth lens **L5**. When the value is within this range, as the development of the ultra-thin and wide-angle lenses, it is beneficial for correcting the off-axis aberration. Preferably, the following condition shall be satisfied, $0.60 \leq (R_9+R_{10})/(R_9-R_{10}) \leq 1.14$.

An on-axis thickness of the fifth lens **L5** is defined as d_9 . The total optical length from the object side surface of the first lens **L1** to the image surface **S1** of the camera optical lens **10** along the optical axis is defined as TTL . The camera optical lens **10** further satisfies the following condition: $0.01 \leq d_9/TTL \leq 0.08$. When the condition is satisfied, it is beneficial for realizing the ultra-thin effect. Preferably, the following condition shall be satisfied, $0.02 \leq d_9/TTL \leq 0.07$.

In the present embodiment, an object side surface of the sixth lens **L6** is convex in the paraxial region, an image side surface of the sixth lens **L6** is concave in the paraxial region, and the sixth lens **L6** has a negative refractive power. In other optional embodiments, the object side surface and the image side surface of the sixth lens **L6** can be arranged as other convex side surface or concave side surface, such as, concave object side surface and convex image side surface and so on. The sixth lens **L6** may have positive refractive power.

The focal length of the camera optical lens **10** is defined as f , and a focal length of the sixth lens **L6** is defined as f_6 . The camera optical lens further satisfies the following condition: $-645.45 \leq f_6/f \leq 138.30$. It is beneficial for realizing the

excellent imaging quality and the lower sensitivity by controlling the refractive power being within reasonable range. Preferably, the following condition shall be satisfied, $-403.41 \leq f_6/f \leq 110.64$.

A central curvature radius of the object side surface of the sixth lens **L6** is defined as R_{11} , and a central curvature radius of the image side surface of the sixth lens **L6** is defined as R_{12} . The camera optical lens further satisfies the following condition: $0.35 \leq (R_{11}+R_{12})/(R_{11}-R_{12}) \leq 64.63$, which specifies a shape of the sixth lens **L6**. When the condition is satisfied, as the development of the ultra-thin and wide-angle lenses, it benefits for correcting the off-axis aberration. Preferably, the following condition shall be satisfied, $0.57 \leq (R_{11}+R_{12})/(R_{11}-R_{12}) \leq 51.70$.

An on-axis thickness of the sixth lens **L6** is defined as d_{11} . The total optical length from the object side surface of the first lens **L1** to the image surface **S1** of the camera optical lens **10** along the optical axis is defined as TTL . The camera optical lens further satisfies the following condition: $0.01 \leq d_{11}/TTL \leq 0.12$, which is beneficial for realizing the ultra-thin effect. Preferably, the following condition shall be satisfied, $0.01 \leq d_{11}/TTL \leq 0.09$.

In the present embodiment, an object side surface of the seventh lens **L7** is convex in the paraxial region, an image side surface of the seventh lens **L7** is convex in the paraxial region, and the seventh lens **L7** has a positive refractive power. In other optional embodiments, the object side surface and the image side surface of the seventh lens **L7** can be arranged as other convex side surface or concave side surface, such as, concave object side surface and concave image side surface and so on. The seventh lens **L7** may have a negative refractive power.

The focal length of the camera optical lens **10** is defined as f , and a focal length of the seventh lens **L7** is defined as f_7 . The camera optical lens **10** further satisfies the following condition: $0.40 \leq f_7/f \leq 1.59$. It is beneficial for realizing the better imaging quality and the lower sensitivity by controlling the refractive power being within reasonable range. Preferably, the following condition shall be satisfied, $0.64 \leq f_7/f \leq 1.27$.

A central curvature radius of the object side surface of the seventh lens **L7** is defined as R_{13} , and a central curvature radius of the image side surface of the seventh lens **L7** is defined as R_{14} . The camera optical lens **10** further satisfies the following condition: $0.38 \leq (R_{13}+R_{14})/(R_{13}-R_{14}) \leq 1.29$, which specifies a shape of the seventh lens **L7**. When the condition is satisfied, as the development of the ultra-thin and wide-angle lenses, it is beneficial for correcting the off-axis aberration. Preferably, the following condition shall be satisfied, $0.60 \leq (R_{13}+R_{14})/(R_{13}-R_{14}) \leq 1.03$.

An on-axis thickness of the seventh lens **L7** is defined as d_{13} . The total optical length from the object side surface of the first lens **L1** to the image surface **S1** of the camera optical lens **10** along the optical axis is defined as TTL . The camera optical lens further satisfies the following condition: $0.05 \leq d_{13}/TTL \leq 0.31$, which is beneficial for realizing the ultra-thin effect. Preferably, the following condition shall be satisfied, $0.08 \leq d_{13}/TTL \leq 0.25$.

In the present embodiment, the object side surface of the eighth lens **L8** is convex in the paraxial region, the image side surface of the eighth lens **L8** is concave in the paraxial region, and the eighth lens **L8** has a negative refractive power. In other optional embodiments, the object side surface and the image side surface of the eighth lens **L8** can be arranged as other convex side surface or concave side

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surface, such as, concave object side surface and convex image side surface and so on. The eighth lens L8 may have a positive refractive power.

The focal length of the camera optical lens 10 is defined as f, and a focal length of the eighth lens L8 is defined as f8. The camera optical lens 10 further satisfies the following condition: $-2.96 \leq f8/f \leq -0.69$. It is beneficial for realizing the better imaging quality and the lower sensitivity by controlling the refractive power being within reasonable range. Preferably, the following condition shall be satisfied, $-1.85 \leq f8/f \leq -0.86$.

A central curvature radius of the object side surface of the eighth lens L8 is defined as R15, and a central curvature radius of the image side surface of the eighth lens L8 is defined as R16. The camera optical lens 10 further satisfies the following condition: $0.75 \leq (R15+R16)/(R15-R16) \leq 3.80$, which specifies a shape of the eighth lens L8. When the condition is satisfied, as the development of the ultra-thin and wide-angle lenses, it is beneficial for correcting the off-axis aberration. Preferably, the following condition shall be satisfied, $1.20 \leq (R15+R16)/(R15-R16) \leq 3.04$.

An on-axis thickness of the eighth lens L8 is defined as d15. The total optical length from the object side surface of the first lens L1 to the image surface S1 of the camera optical lens 10 along the optical axis is defined as TTL. The camera optical lens further satisfies the following condition: $0.03 \leq d15/TTL \leq 0.15$, which is beneficial for realizing the ultra-thin effect. Preferably, the following condition shall be satisfied, $0.05 \leq d15/TTL \leq 0.12$.

In the present embodiment, an image height of the camera optical lens 10 is defined as IH. The total optical length from the object side surface of the first lens L1 to the image surface S1 of the camera optical lens 10 along an optical axis is defined as TTL. The camera optical lens 10 further satisfies the following condition: $TTL/IH \leq 2.91$, thereby achieving the ultra-thin performance. Preferably, the following condition shall be satisfied, $TTL/IH \leq 2.83$.

In the present embodiment, an F number (FNO) of the camera optical lens 10 is smaller than or equal to 2.68, thereby achieving the large aperture and good imaging performance. Preferably, the FNO of the camera optical lens 10 is smaller than or equal to 2.63.

When the above conditions are satisfied, which makes it is possible that the camera optical lens has excellent optical performances, and meanwhile can meet design requirements of large aperture. According the characteristics of the camera optical lens 10, it is particularly suitable for a mobile camera lens component and a WEB camera lens composed of high pixel CCD, CMOS.

The following examples will be used to describe the camera optical lens 10 of the present invention. The symbols recorded in each example will be described as follows. The focal length, on-axis distance, central curvature radius, on-axis thickness, inflexion point position, and arrest point position are all in units of mm.

TTL: the total optical length from the object side surface of the first lens L1 to the image surface S1 of the camera optical lens 10 along the optical axis, the unit of TTL is mm.

F number (FNO): a ratio of an effective focal length of the camera optical lens 10 to an entrance pupil diameter (ENPD).

Preferably, inflexion points and/or arrest points can also be arranged on the object side surface and/or image side surface of the lens, so that the demand for high quality imaging can be satisfied, the description below can be referred for specific implementable scheme.

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The design information of the camera optical lens 10 in Embodiment 1 of the present invention is shown in the tables 1 and 2.

TABLE 1

	R	d	nd	vd
S1	∞	d0=	-3.079	
R1	-7.479	d1=	1.306 nd1	1.5444 v1 55.82
R2	1.935	d2=	0.455	
R3	2.339	d3=	0.332 nd2	1.6610 v2 20.53
R4	4.207	d4=	0.234	
R5	89.494	d5=	0.576 nd3	1.6700 v3 19.39
R6	109.026	d6=	0.138	
R7	3.749	d7=	1.151 nd4	1.5444 v4 55.82
R8	-2.873	d8=	0.062	
R9	-60.406	d9=	0.219 nd5	1.6700 v5 19.39
R10	8.814	d10=	0.737	
R11	13.005	d11=	0.763 nd6	1.6359 v6 23.82
R12	12.415	d12=	0.050	
R13	12.224	d13=	1.029 nd7	1.5444 v7 55.82
R14	-1.72	d14=	0.489	
R15	9.977	d15=	0.559 nd8	1.6400 v8 23.54
R16	1.997	d16=	1.285	
R17	∞	d17=	0.275 ndg	1.5168 vg 64.20
R18	∞	d18=	0.110	

where, the meaning of the various symbols is as follows.
S1: aperture;

R: curvature radius of an optical surface, a central curvature radius for a lens;

R1: central curvature radius of the object side surface of the first lens L1;

R2: central curvature radius of the image side surface of the first lens L1;

R3: central curvature radius of the object side surface of the second lens L2;

R4: central curvature radius of the image side surface of the second lens L2;

R5: central curvature radius of the object side surface of the third lens L3;

R6: central curvature radius of the image side surface of the third lens L3;

R7: central curvature radius of the object side surface of the fourth lens L4;

R8: central curvature radius of the image side surface of the fourth lens L4;

R9: central curvature radius of the object side surface of the fifth lens L5;

R10: central curvature radius of the image side surface of the fifth lens L5;

R11: central curvature radius of the object side surface of the sixth lens L6;

R12: central curvature radius of the image side surface of the sixth lens L6;

R13: central curvature radius of the object side surface of the seventh lens L7;

R14: central curvature radius of the image side surface of the seventh lens L7;

R15: central curvature radius of the object side surface of the eighth lens L8;

R16: central curvature radius of the image side surface of the eighth lens L8;

R17: central curvature radius of an object side surface of the optical filter GF;

R18: curvature radius of an image side surface of the optical filter GF;

d: on-axis thickness of a lens and an on-axis distance between lenses;

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d0: on-axis distance from the aperture S1 to the object side surface of the first lens L1;
d1: on-axis thickness of the first lens L1;
d2: on-axis distance from the image side surface of the first lens L1 to the object side surface of the second lens L2;
d3: on-axis thickness of the second lens L2;
d4: on-axis distance from the image side surface of the second lens L2 to the object side surface of the third lens L3;
d5: on-axis thickness of the third lens L3;
d6: on-axis distance from the image side surface of the third lens L3 to the object side surface of the fourth lens L4;
d7: on-axis thickness of the fourth lens L4;
d8: on-axis distance from the image side surface of the fourth lens L4 to the object side surface of the fifth lens L5;
d9: on-axis thickness of the fifth lens L5;
d10: on-axis distance from the image side surface of the fifth lens L5 to the object side surface of the sixth lens L6;
d11: on-axis thickness of the sixth lens L6;
d12: on-axis distance from the image side surface of the sixth lens L6 to the object side surface of the seventh lens L7;
d13: on-axis thickness of the seventh lens L7;
d14: on-axis distance from the image side surface of the seventh lens L7 to the object side surface of the eighth lens L8;

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d15: on-axis thickness of the eighth lens L8;
d16: on-axis distance from the image side surface of the eighth lens L8 to the object side surface of the optical filter GF;
d17: on-axis thickness of the optical filter GF;
d18: on-axis distance from the image side surface of the optical filter GF to the image surface;
nd: refractive index of d line (d-line is green light with a wavelength of 550 nm);
nd1: refractive index of d line of the first lens L1;
nd2: refractive index of d line of the second lens L2;
nd3: refractive index of d line of the third lens L3;
nd4: refractive index of d line of the fourth lens L4;
nd5: refractive index of d line of the fifth lens L5;
nd6: refractive index of d line of the sixth lens L6;
nd7: refractive index of d line of the seventh lens L7;
nd8: refractive index of d line of the eighth lens L8;
ndg: refractive index of d line of the optical filter GF;
vd: abbe number;
v1: abbe number of the first lens L1;
v2: abbe number of the second lens L2;
v3: abbe number of the third lens L3;
v4: abbe number of the fourth lens L4;
v5: abbe number of the fifth lens L5;
v6: abbe number of the sixth lens L6;
v7: abbe number of the seventh lens L7;
v8: abbe number of the eighth lens L8;
vg: abbe number of the optical filter GF;

Table 2 shows the aspherical surface data of the camera optical lens 10 in Embodiment 1 of the present invention.

TABLE 2

Conic coefficient		Aspheric surface coefficients				
	k	A4	A6	A8	A10	A12
R1	2.2006E+00	4.3507E-02	-1.2003E-02	2.7849E-03	-4.8370E-04	6.0133E-05
R2	8.3815E-01	8.6747E-02	-2.5639E-02	-2.8534E-04	3.5873E-03	-1.0567E-03
R3	1.2495E+00	-1.9668E-02	-1.3041E-02	1.3000E-03	1.2574E-03	7.4346E-04
R4	-1.9969E+00	-1.6163E-03	6.5094E-03	2.1667E-03	1.9578E-03	1.5277E-03
R5	1.7941E+03	2.7481E-03	-3.8342E-03	-9.0799E-04	2.7356E-04	1.0816E-03
R6	4.7445E+03	1.3601E-04	-1.2405E-02	5.3279E-03	5.7504E-03	4.9519E-03
R7	1.0753E+01	-1.3083E-02	-2.5558E-02	6.0440E-04	-1.7726E-02	-7.6010E-03
R8	4.7219E+00	-1.1832E-01	7.4041E-02	-8.3437E-02	7.2312E-02	-5.5362E-02
R9	2.7091E+03	-2.1282E-01	7.6881E-02	-9.1082E-02	5.8430E-02	-3.5492E-02
R10	-6.0510E+01	-8.7439E-02	3.7036E-02	-1.4442E-02	7.0042E-03	-2.1623E-03
R11	-3.3700E+02	-2.3259E-03	3.1154E-04	8.6345E-06	-6.7870E-06	-1.0480E-06
R12	-1.1425E+02	-6.4922E-03	-1.0148E-03	-1.3697E-04	-6.5904E-06	5.2212E-07
R13	1.8623E+01	-1.3940E-02	-5.0040E-04	6.8796E-04	-2.2121E-04	-1.3589E-05
R14	-2.0315E+00	3.3249E-02	-1.7572E-02	6.1337E-03	-1.3991E-03	2.2819E-04
R15	3.0701E+00	8.9458E-03	-8.4414E-03	1.3469E-03	-9.8952E-05	-3.7634E-06
R16	-4.4084E+00	-6.3795E-03	-6.2535E-04	1.7896E-04	-2.1204E-05	1.4138E-06

Conic coefficient		Aspheric surface coefficients			
	k	A14	A16	A18	A20
R1	2.2006E+00	-5.1432E-06	2.8260E-07	-8.7977E-09	1.1484E-10
R2	8.3815E-01	-6.3920E-04	-3.2825E-04	-1.4503E-04	5.9978E-05
R3	1.2495E+00	1.4804E-04	-6.4878E-05	-8.3018E-05	-2.5939E-05
R4	-1.9969E+00	5.9323E-04	2.2354E-05	-1.9602E-04	-1.6884E-04
R5	1.7941E+03	1.6774E-04	-1.5055E-04	-7.7486E-05	1.0732E-04
R6	4.7445E+03	8.2284E-04	-5.2901E-03	-5.5529E-03	6.8382E-03
R7	1.0753E+01	1.0705E-02	-1.1613E-02	1.5403E-02	-4.1957E-02
R8	4.7219E+00	3.0551E-02	-4.7274E-03	-8.8967E-04	-4.0382E-04
R9	2.7091E+03	1.0974E-02	-1.4727E-03	7.8710E-04	1.9829E-04
R10	-6.0510E+01	3.2233E-04	6.7714E-06	2.7774E-05	-1.1130E-05
R11	-3.3700E+02	2.0528E-08	2.6469E-08	4.8590E-09	-7.0569E-10
R12	-1.1425E+02	1.3088E-07	8.5478E-09	-1.3853E-10	-1.1202E-10
R13	1.8623E+01	1.8388E-05	-3.4453E-06	2.1272E-07	-1.8546E-09
R14	-2.0315E+00	-2.5074E-05	1.4677E-06	-3.7363E-08	4.9811E-10
R15	3.0701E+00	-9.7651E-08	4.7247E-08	1.6278E-08	-3.2831E-09
R16	-4.4084E+00	-5.5540E-08	1.0955E-09	-9.2673E-12	1.2493E-13

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For convenience, an aspheric surface of each lens surface uses the aspheric surfaces shown in the below condition (1). However, the present invention is not limited to the aspherical polynomials form shown in the condition (1).

$$z=(cr^2)/\{1+[1-(k+1)(c^2r^2)]^{1/2}\}+A4r^4+A6r^6+A8r^8+A10r^{10}+A12r^{12}+A14r^{14}+A16r^{16}+A18r^{18}+A20r^{20} \quad (1)$$

Where, K is a conic coefficient, A4, A6, A8, A10, A12, A14, A16, A18, A20 are aspheric surface coefficients. c is the curvature at the center of the optical surface. r is a vertical distance between a point on an aspherical curve and the optic axis, and z is an aspherical depth (a vertical distance between a point on an aspherical surface, having a distance of r from the optic axis, and a surface tangent to a vertex of the aspherical surface on the optic axis).

Table 3 and Table 4 show design data of inflexion points and arrest points of respective lens in the camera optical lens 10 according to Embodiment 1 of the present invention. P1R1 and P1R2 represent the object side surface and the image side surface of the first lens L1, P2R1 and P2R2 represent the object side surface and the image side surface of the second lens L2, P3R1 and P3R2 represent the object side surface and the image side surface of the third lens L3, P4R1 and P4R2 represent the object side surface and the image side surface of the fourth lens L4, P5R1 and P5R2 represent the object side surface and the image side surface of the fifth lens L5, P6R1 and P6R2 represent the object side surface and the image side surface of the sixth lens L6, and P7R1 and P7R2 represent the object side surface and the image side surface of the seventh lens L7, and P8R1 and P8R2 represent the object side surface and the image side surface of the eighth lens L8. The data in the column named “inflexion point position” refers to vertical distances from inflexion points arranged on each lens surface to the optical axis of the camera optical lens 10. The data in the column named “arrest point position” refers to vertical distances from arrest points arranged on each lens surface to the optical axis of the camera optical lens 10.

TABLE 3

	Number of inflexion points	Inflexion point position 1	Inflexion point position 2
P1R1	2	0.575	2.785
P1R2	0	/	/
P2R1	0	/	/
P2R2	0	/	/
P3R1	2	0.725	0.895
P3R2	2	0.445	0.705
P4R1	1	0.785	/
P4R2	0	/	/
P5R1	0	/	/
P5R2	2	0.335	1.225
P6R1	2	0.935	2.305
P6R2	1	0.725	/
P7R1	1	0.745	/
P7R2	2	1.695	1.985
P8R1	1	1.065	/
P8R2	1	1.265	/

TABLE 4

	Number of arrest points	Arrest point position 1
P1R1	1	1.115
P1R2	0	/
P2R1	0	/
P2R2	0	/
P3R1	0	/

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TABLE 4-continued

	Number of arrest points	Arrest point position 1
P3R2	0	/
P4R1	0	/
P4R2	0	/
P5R1	0	/
P5R2	1	0.595
P6R1	1	2.065
P6R2	1	1.215
P7R1	1	1.305
P7R2	0	/
P8R1	1	1.595
P8R2	1	2.745

FIG. 2 and FIG. 3 respectively illustrate a longitudinal aberration and a lateral color of light with wavelengths of 656 nm, 588 nm, 546 nm, 486 nm and 436 nm after passing the camera optical lens 10 according to Embodiment 1. FIG. 4 illustrates a field curvature and a distortion of light with a wavelength of 546 nm after passing the camera optical lens 10 according to Embodiment 1, in which a field curvature S is a field curvature in a sagittal direction and T is a field curvature in a tangential direction.

Table 13 shows various values of Embodiments 1, 2 and 3 and values corresponding to parameters which are specified in the above conditions.

As shown in Table 13, Embodiment 1 satisfies the above conditions.

In the present embodiment, the entrance pupil diameter (ENPD) of the camera optical lens 10 is 1.029 mm. The image height of 1.0H is 5.150 mm. The FOV is 133.80°. Thus, the camera optical lens 10 satisfies design requirements of wide-angle while the on-axis and off-axis aberrations are sufficiently corrected, thereby achieving excellent optical characteristics.

Embodiment 2

Embodiment 2 is basically the same as Embodiment 1, the meaning of its symbols is the same as that of Embodiment 1, in the following, only the differences are listed.

A sixth lens L6 has an image side surface being convex in the paraxial region. A second lens L2 has a negative refractive power, and the sixth lens L6 has a positive refractive power.

FIG. 5 shows a schematic diagram of a structure of a camera optical lens 20 according to Embodiment 2 of the present invention. Table 5 and table 6 show the design data of a camera optical lens 20 in Embodiment 2 of the present invention.

TABLE 5

	R	d	nd	vd
S1	∞	d0=	-3.007	
R1	-11.509	d1=	0.702	nd1
R2	5.422	d2=	1.196	
R3	8.727	d3=	0.154	nd2
R4	5.591	d4=	0.025	
R5	4.244	d5=	0.492	nd3
R6	11.250	d6=	0.402	
R7	6.783	d7=	0.908	nd4
R8	-3.602	d8=	0.215	
R9	-194.997	d9=	0.569	nd5
R10	5.349	d10=	0.368	
R11	3032.878	d11=	0.224	nd6
R12	-203.353	d12=	0.151	
R13	15.692	d13=	2.158	nd7
R14	-1.508	d14=	0.214	

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TABLE 5-continued

	R	d	nd	vd
R15	5.023	d15=	1.038	nd8
R16	1.358	d16=	1.070	1.6400 v8
R17	∞	d17=	0.275	ndg
R18	∞	d18=	0.267	1.5168 vg
				64.20

Table 6 shows aspherical surface data of each lens of the camera optical lens **20** in Embodiment 2 of the present invention.

TABLE 6

	Conic coefficient	Aspheric surface coefficients				
	k	A4	A6	A8	A10	A12
R1	2.9319E+00	4.4693E-02	-1.2190E-02	2.7871E-03	-4.8355E-04	6.0125E-05
R2	1.4672E+00	5.9052E-02	-8.9279E-03	-8.6529E-04	1.3351E-03	-3.6597E-04
R3	-2.2192E+00	4.5232E-03	-3.9953E-03	-8.7220E-04	-2.3449E-04	3.0081E-04
R4	-3.8845E-01	-1.3104E-03	-9.6556E-04	-6.5828E-04	-1.0618E-04	2.5244E-05
R5	-2.9774E+00	-4.8388E-03	-7.4631E-04	2.2041E-04	-8.5287E-06	-2.5444E-05
R6	2.4935E+01	-3.2385E-04	-1.7779E-03	1.4050E-03	-7.5135E-04	1.0958E-03
R7	1.2823E+01	7.5380E-03	-9.8230E-03	-5.7288E-04	-1.7047E-04	-8.2258E-04
R8	5.4528E+00	-5.0199E-02	5.2900E-02	-7.3638E-02	7.5927E-02	-5.8918E-02
R9	-1.2802E-01	7.6469E-02	-8.1703E-02	6.1748E-02	-3.4434E-02	1.0806E-02
R10	1.3619E-01	-7.0012E-02	3.6847E-02	-1.7941E-02	6.7417E-03	-1.9267E-03
R11	-1.0306E+04	1.4461E-03	1.3523E-04	6.1830E-06	-6.6414E-06	-1.3678E-06
R12	-1.3193E-03	-2.5644E-04	-6.1056E-05	-2.4193E-06	8.1661E-07	2.5493E-07
R13	3.9649E+01	-9.7647E-03	3.0673E-04	6.4457E-04	-2.2965E-04	-1.4259E-05
R14	-1.4395E+00	3.3973E-02	-1.8081E-02	6.0170E-03	-1.3941E-03	2.3043E-04
R15	-4.4518E+01	-1.9262E-03	-4.4021E-03	9.4866E-04	-9.3261E-05	3.5013E-06
R16	-4.6732E+00	-5.3363E-03	-5.1815E-04	1.7981E-04	-2.1576E-05	1.4195E-06

	Conic coefficient	Aspheric surface coefficients			
	k	A14	A16	A18	A20
R1	2.9319E+00	-5.1403E-06	2.8294E-07	-8.7611E-09	1.1017E-10
R2	1.4672E+00	2.9539E-05	4.8064E-08	4.6258E-08	3.1686E-08
R3	-2.2192E+00	-6.0361E-05	-2.9983E-06	-4.1820E-07	4.5525E-08
R4	-3.8845E-01	2.2003E-05	6.5781E-06	4.7688E-07	4.8645E-08
R5	-2.9774E+00	-3.6008E-06	1.0275E-05	9.5044E-06	4.5146E-06
R6	2.4935E+01	-4.2880E-04	-9.3349E-05	-1.3816E-05	6.4371E-05
R7	1.2823E+01	-1.2892E-03	-1.3751E-03	-9.7902E-04	-3.9296E-04
R8	5.4528E+00	2.6758E-02	-6.5353E-03	5.6999E-04	-1.7779E-04
R9	-1.2802E-01	-2.1566E-03	1.3914E-04	3.2726E-06	0.0000E+00
R10	1.3619E-01	3.6155E-04	-4.0079E-05	1.8384E-06	4.8191E-08
R11	-1.0306E+04	-2.5829E-08	3.2128E-08	9.1544E-09	-1.1585E-10
R12	-1.3193E-03	4.6802E-08	8.4870E-09	1.5591E-09	0.0000E+00
R13	3.9649E+01	1.8293E-05	-3.4656E-06	2.1039E-07	-1.4836E-09
R14	-1.4395E+00	-2.4780E-05	1.4869E-06	-3.8382E-08	-1.0292E-10
R15	-4.4518E+01	7.3994E-08	-8.7602E-09	1.8618E-10	-1.0705E-12
R16	-4.6732E+00	-5.4568E-08	1.1092E-09	-1.0213E-11	5.5522E-14

Table 7 and table 8 show design data of inflexion points and arrest points of respective lens in the camera optical lens **20** according to Embodiment 2 of the present invention.

TABLE 7

	Number of inflexion points	Inflexion point position1
P1R1	1	0.435
P1R2	0	/
P2R1	1	1.015
P2R2	0	/
P3R1	0	/
P3R2	0	/
P4R1	1	0.855
P4R2	0	/
P5R1	0	/
P5R2	1	0.585
P6R1	0	/
P6R2	1	1.885

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TABLE 7-continued

	Number of inflexion points	Inflexion point position1
P7R1	1	0.915
P7R2	0	/
P8R1	1	0.835
P8R2	1	1.195

TABLE 8

	Number of arrest points	Arrest point position1
P1R1	1	0.785
P1R2	0	/
P2R1	1	1.445
P2R2	0	/
P3R1	0	/
P3R2	0	/
P4R1	0	/
P4R2	0	/
P5R1	0	/
P5R2	1	1.215
P6R1	0	/
P6R2	1	2.095
P7R1	1	1.645
P7R2	0	/
P8R1	1	1.525
P8R2	1	3.295

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FIG. 6 and FIG. 7 respectively illustrate a longitudinal aberration and a lateral color of light with wavelengths of 656 nm, 588 nm, 546 nm, 486 nm and 436 nm after passing the camera optical lens **20** according to Embodiment 2. FIG. 8 illustrates a field curvature and a distortion of light with a wavelength of 546 nm after passing the camera optical lens **10** according to Embodiment 2, in which a field curvature S is a field curvature in a sagittal direction and T is a field curvature in a tangential direction.

As shown in Table 13, Embodiment 2 satisfies the various conditions.

In the present embodiment, an entrance pupil diameter (ENPD) of the camera optical lens **20** is 1.376 mm. An image height of 1.0H is 4.350 mm. An FOV is 115.80°. Thus, the camera optical lens **20** satisfies design requirements of wide-angle while on-axis and off-axis aberrations are sufficiently corrected, thereby achieving excellent optical characteristics.

Embodiment 3

Embodiment 3 is basically the same as Embodiment 1 and involves symbols having the same meanings as Embodiment 1, and only differences therebetween will be described in the following.

A sixth lens L6 has an object side surface being concave in a paraxial region. A second lens L2 has a negative refractive power.

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FIG. 9 shows a schematic diagram of a structure of a camera optical lens **30** according to Embodiment 3 of the present invention.

Tables 9 and 10 show design data of a camera optical lens **30** in Embodiment 3 of the present invention.

TABLE 9

	R	d	nd	vd
10 S1	∞	d0=	-2.651	
R1	-16.078	d1=	0.542	nd1
R2	8.795	d2=	1.026	
R3	20.565	d3=	0.105	nd2
R4	4.709	d4=	0.149	
R5	3.709	d5=	0.393	nd3
15 R6	11.956	d6=	0.399	
R7	7.051	d7=	0.839	nd4
R8	-3.737	d8=	0.248	
R9	-48.808	d9=	0.534	nd5
R10	5.858	d10=	0.467	
R11	-195.438	d11=	0.151	nd6
R12	33.322	d12=	0.134	
20 R13	19.3	d13=	2.024	nd7
R14	-1.474	d14=	0.265	
R15	2.594	d15=	0.775	nd8
R16	1.127	d16=	1.271	
R17	∞	d17=	0.275	ndg
25 R18	∞	d18=	0.228	

Table 10 shows aspherical surface data of each lens of the camera optical lens **30** in Embodiment 3 of the present invention.

TABLE 10

	Conic coefficient	Aspheric surface coefficients				
	k	A4	A6	A8	A10	A12
R1	-6.3522E+00	4.6150E-02	-1.2198E-02	2.7802E-03	-4.8442E-04	6.0055E-05
R2	-5.6062E-01	5.8391E-02	-9.0126E-03	-8.6214E-04	1.3157E-03	-3.6905E-04
R3	-8.0782E+00	3.8593E-03	-4.0448E-03	-8.4274E-04	-2.0313E-04	3.2121E-04
R4	-7.2734E-03	-3.9581E-04	-9.0565E-04	-8.6067E-04	-1.7313E-04	2.0948E-05
R5	-2.5867E+00	-4.2515E-03	-1.0778E-03	1.1159E-04	-5.1422E-05	-5.8132E-05
R6	5.2424E+00	-1.6705E-03	-1.5913E-03	1.4836E-03	-6.9072E-04	1.1502E-03
R7	7.0023E+00	6.0446E-03	-9.4580E-03	4.8573E-04	6.8203E-04	-8.8034E-04
R8	5.4138E+00	-5.0595E-02	5.2567E-02	-7.3784E-02	7.5735E-02	-5.9099E-02
R9	1.4075E+03	-1.2886E-01	7.6944E-02	-8.0810E-02	6.2090E-02	-3.4368E-02
R10	-5.6014E-01	-7.0292E-02	3.6548E-02	-1.7971E-02	6.7567E-03	-1.9190E-03
R11	7.1278E+03	1.0555E-03	-9.2075E-06	-1.8601E-05	-9.3365E-06	-1.7253E-06
R12	-1.4733E+03	-1.8166E-03	-2.1906E-04	-4.9239E-05	-1.6305E-06	7.3820E-07
R13	3.3420E+01	-1.0587E-02	3.3402E-04	6.3327E-04	-2.3107E-04	-1.3901E-05
R14	-1.3388E+00	3.3186E-02	-1.8141E-02	6.0078E-03	-1.3953E-03	2.3025E-04
R15	-9.3147E+00	-1.0632E-03	-4.3038E-03	9.5622E-04	-9.2907E-05	3.4961E-06
R16	-3.3287E+00	-7.5519E-03	-3.2272E-04	1.7826E-04	-2.1879E-05	1.4181E-06

	Conic coefficient	Aspheric surface coefficients			
	k	A14	A16	A18	A20
R1	-6.3522E+00	-5.1432E-06	2.8314E-07	-8.6985E-09	1.1885E-10
R2	-5.6062E-01	2.9665E-05	1.5567E-07	3.6715E-08	8.2876E-09
R3	-8.0782E+00	-5.0664E-05	2.9316E-07	2.9594E-07	4.6464E-08
R4	-7.2734E-03	2.8467E-05	1.1926E-05	3.3503E-06	7.8458E-07
R5	-2.5867E+00	-3.0504E-05	-9.7648E-06	-4.3155E-06	-4.9686E-06
R6	5.2424E+00	-4.5504E-04	-1.3116E-04	-4.9732E-05	3.7482E-05
R7	7.0023E+00	-2.3674E-03	-2.7953E-03	-1.1301E-03	3.4082E-03
R8	5.4138E+00	2.6711E-02	-6.4455E-03	6.3245E-04	-1.6264E-04
R9	1.4075E+03	1.0844E-02	-2.1109E-03	1.7495E-04	3.1914E-05
R10	-5.6014E-01	3.6380E-04	-3.9592E-05	1.8675E-06	-7.4266E-10
R11	7.1278E+03	-9.3710E-08	2.4944E-08	1.2395E-08	2.9124E-09
R12	-1.4733E+03	2.2762E-07	3.9937E-08	5.5795E-09	2.1506E-10
R13	3.3420E+01	1.8444E-05	-3.4339E-06	2.1514E-07	-9.8398E-10
R14	-1.3388E+00	-2.4806E-05	1.4843E-06	-3.8612E-08	-9.9278E-11
R15	-9.3147E+00	7.1884E-08	-8.9357E-09	1.8013E-10	-5.0302E-13
R16	-3.3287E+00	-5.3961E-08	1.1285E-09	-1.0053E-11	-1.7644E-14

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Table 11 and table 12 show Embodiment 3 design data of inflexion points and arrest points of respective lens in the camera optical lens **30** according to Embodiment 3 of the present invention.

TABLE 11

	Number of inflexion points	Inflexion point position1	Inflexion point position2
P1R1	1	0.355	/
P1R2	0	/	/
P2R1	1	0.865	/
P2R2	0	/	/
P3R1	1	1.145	/
P3R2	0	/	/
P4R1	0	/	/
P4R2	0	/	/
P5R1	0	/	/
P5R2	1	0.545	/
P6R1	2	0.705	1.245
P6R2	2	0.725	1.925
P7R1	1	0.695	/
P7R2	0	/	/
P8R1	1	1.045	/
P8R2	1	1.205	/

TABLE 12

	Number of arrest points	Arrest point position1
P1R1	1	0.625
P1R2	0	/
P2R1	1	1.255
P2R2	0	/
P3R1	0	/
P3R2	0	/
P4R1	0	/
P4R2	0	/
P5R1	0	/
P5R2	1	1.085
P6R1	0	/
P6R2	1	1.275
P7R1	1	1.275
P7R2	0	/
P8R1	1	2.155
P8R2	0	/

FIG. **10** and FIG. **11** respectively illustrate a longitudinal aberration and a lateral color of light with wavelengths of 656 nm, 588 nm, 546 nm, 486 nm and 436 nm after passing the camera optical lens **30** according to Embodiment 3. FIG. **12** illustrates a field curvature and a distortion of light with a wavelength of 546 nm after passing the camera optical lens **30** according to Embodiment 3, in which a field curvature S is a field curvature in a sagittal direction and T is a field curvature in a tangential direction.

Table 13 in the following lists values corresponding to the respective conditions. In the present Embodiment 3 in order to satisfy the above conditions.

In the present embodiment, an entrance pupil diameter (ENPD) of the camera optical lens **30** is 1.411 mm. An image height of 1.0H is 3.540 mm. An FOV is 100.00°. Thus, the camera optical lens **30** satisfies design requirements of wide-angle while on-axis and off-axis aberrations are sufficiently corrected, thereby achieving excellent optical characteristics.

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TABLE 13

Parameters and conditions	Embodiment 1	Embodiment 2	Embodiment 3
5 FOV	133.80°	115.80°	100.00°
f1/f	-1.00	-2.11	-3.20
R15/R16	5.00	3.70	2.30
f	2.675	3.145	3.225
f1	-2.680	-6.644	-10.319
f2	7.352	-26.898	-10.389
10 f3	727.994	9.773	7.780
f4	3.170	4.441	4.594
f5	-11.329	-7.668	-7.682
f6	-863.295	289.970	-43.304
f7	2.832	2.632	2.594
f8	-3.957	-3.240	-3.894
15 FNO	2.60	2.29	2.29
TTL	9.770	10.428	9.825
IH	5.150	4.350	3.540

It is to be understood, however, that even though numerous characteristics and advantages of the present exemplary embodiments have been set forth in the foregoing description, together with details of the structures and functions of the embodiments, the disclosure is illustrative only, and changes may be made in detail, especially in matters of shape, size, and arrangement of parts within the principles of the invention to the full extent indicated by the broad general meaning of the terms where the appended claims are expressed.

What is claimed is:

1. A camera optical lens, comprising, from an object side to an image side in sequence: a first lens having a negative refractive power, a second lens, a third lens, a fourth lens, a fifth lens, a sixth lens, a seventh lens, and an eighth lens; the first lens has an object side surface being concave in a paraxial region and an image side surface being concave in the paraxial region; wherein the camera optical lens satisfies the following conditions:

100.00°≤FOV≤135.00°;
3.20≤f1/f≤-1.00;
2.30≤R15/R16≤5.00;
0.15≤(R1+R2)/(R1-R2)≤0.88; and
0.03≤d1/TTL≤0.20;

where,

FOV: a field of view of the camera optical lens in a diagonal direction;

f: a focal length of the camera optical lens;

f1: a focal length of the first lens;

R1: a central curvature radius of the object side surface of the first lens;

R2: a central curvature radius of the image side surface of the first lens;

d1: an on-axis thickness of the first lens;

TTL: a total optical length from the object side surface of the first lens of the camera optical lens to an image surface of the camera optical lens along an optical axis;

R15: a central curvature radius of an object side surface of the eighth lens; and

R16: a central curvature radius of an image side surface of the eighth lens.

2. The camera optical lens according to claim 1, wherein, the fifth lens has a negative refractive power; wherein the camera optical lens satisfies the following condition:

-4.50≤f5/f≤-2.00;

f5: a focal length of the fifth lens.

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3. The camera optical lens according to claim 1 further satisfying the following conditions:

$$0.23 \leq (R1+R2)/(R1-R2) \leq 0.71; \text{ and} \\ 0.04 \leq d1/TTL \leq 0.16.$$

4. The camera optical lens according to claim 1, wherein, the second lens has an object side surface being convex in a paraxial region, an image side surface being concave in the paraxial region; the camera optical lens further satisfies the following conditions:

$$-17.11 \leq f2/f \leq 4.12; \\ -7.01 \leq (R3+R4)/(R3-R4) \leq 6.85; \text{ and} \\ 0.01 \leq d3/TTL \leq 0.05;$$

where,

f2: a focal length of the second lens;

R3: a central curvature radius of the object side surface of the second lens;

R4: a central curvature radius of the image side surface of the second lens; and

d3: an on-axis thickness of the second lens.

5. The camera optical lens according to claim 4 further satisfying the following conditions:

$$-10.69 \leq f2/f \leq 3.30; \\ -4.38 \leq (R3+R4)/(R3-R4) \leq 5.48; \text{ and} \\ 0.01 \leq d3/TTL \leq 0.04.$$

6. The camera optical lens according to claim 1, wherein, the third lens has an object side surface being convex in a paraxial region, an image side surface being concave in the paraxial region;

the third lens has a positive refractive power;

the camera optical lens further satisfies the following conditions:

$$1.21 \leq f3/f \leq 408.22; \\ -20.33 \leq (R5+R6)/(R5-R6) \leq -1.27; \text{ and} \\ 0.02 \leq d5/TTL \leq 0.09;$$

where,

f3: a focal length of the third lens;

R5: a central curvature radius of the object side surface of the third lens;

R6: a central curvature radius of the image side surface of the third lens; and

d5: an on-axis thickness of the third lens.

7. The camera optical lens according to claim 6 further satisfying the following conditions:

$$1.93 \leq f3/f \leq 326.58; \\ -12.70 \leq (R5+R6)/(R5-R6) \leq -1.58; \text{ and} \\ 0.03 \leq d5/TTL \leq 0.07.$$

8. The camera optical lens according to claim 1, wherein, the fourth lens has an object side surface being convex in a paraxial region, an image side surface being convex in the paraxial region;

the fourth lens has a positive refractive power;

the camera optical lens further satisfies the following conditions:

$$0.59 \leq f4/f \leq 2.14; \\ 0.07 \leq (R7+R8)/(R7-R8) \leq 0.46; \text{ and} \\ 0.04 \leq d7/TTL \leq 0.18;$$

where,

f4: a focal length of the fourth lens;

R7: a central curvature radius of the object side surface of the fourth lens;

R8: a central curvature radius of the image side surface of the fourth lens; and

d7: an on-axis thickness of the fourth lens.

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9. The camera optical lens according to claim 8 further satisfying the following conditions:

$$0.95 \leq f4/f \leq 1.71; \\ 0.11 \leq (R7+R8)/(R7-R8) \leq 0.37; \text{ and} \\ 0.07 \leq d7/TTL \leq 0.14.$$

10. The camera optical lens according to claim 1, wherein, the fifth lens has an object side surface being concave in a paraxial region and an image side surface being concave in the paraxial region;

the camera optical lens further satisfies the following conditions:

$$0.37 \leq (R9+R10)/(R9-R10) \leq 1.42; \text{ and} \\ 0.01 \leq d9/TTL \leq 0.08;$$

where,

R9: a central curvature radius of the object side surface of the fifth lens;

R10: a central curvature radius of the image side surface of the fifth lens; and

d9: an on-axis thickness of the fifth lens.

11. The camera optical lens according to claim 10 further satisfying the following conditions:

$$0.60 \leq (R9+R10)/(R9-R10) \leq 1.14; \text{ and} \\ 0.02 \leq d9/TTL \leq 0.07.$$

12. The camera optical lens according to claim 1 further satisfying the following conditions:

$$-645.45 \leq f6/f \leq 138.30; \\ 0.35 \leq (R11+R12)/(R11-R12) \leq 64.63; \text{ and} \\ 0.01 \leq d11/TTL \leq 0.12;$$

where,

f6: a focal length of the sixth lens;

R11: a central curvature radius of an object side surface of the sixth lens;

R12: a central curvature radius of an image side surface of the sixth lens; and

d11: an on-axis thickness of the sixth lens.

13. The camera optical lens according to claim 12 further satisfying the following conditions:

$$-403.41 \leq f6/f \leq 110.64; \\ 0.57 \leq (R11+R12)/(R11-R12) \leq 51.70; \text{ and} \\ 0.01 \leq d11/TTL \leq 0.09.$$

14. The camera optical lens according to claim 1, wherein, the seventh lens has an object side surface being convex in a paraxial region and an image side surface being convex in a paraxial region;

the seventh lens has a positive refractive power;

the camera optical lens further satisfies the following conditions:

$$0.40 \leq f7/f \leq 1.59; \\ 0.38 \leq (R13+R14)/(R13-R14) \leq 1.29; \text{ and} \\ 0.05 \leq d13/TTL \leq 0.31;$$

where,

f7: a focal length of the seventh lens;

R13: a central curvature radius of the object side surface of the seventh lens;

R14: a central curvature radius of the image side surface of the seventh lens; and

d13: an on-axis thickness of the seventh lens.

15. The camera optical lens according to claim 14 further satisfying the following conditions:

$$0.64 \leq f7/f \leq 1.27; \\ 0.60 \leq (R13+R14)/(R13-R14) \leq 1.03; \text{ and} \\ 0.08 \leq d13/TTL \leq 0.25.$$

16. The camera optical lens according to claim 1, where, the object side surface of the eighth lens is convex in a paraxial region and the image side surface of the eighth lens is concave in the paraxial region;

the eighth lens has a negative refractive power;
the camera optical lens further satisfies the following
conditions:

$$-2.96 \leq f_8/f \leq -0.69;$$

$$0.75 \leq (R15+R16)/(R15-R16) \leq 3.80; \text{ and} \quad 5$$

$$0.03 \leq d15/TTL \leq 0.15;$$

wherein,

f_8 : a focal length of the eighth lens; and

$d15$: an on-axis thickness of the eighth lens.

17. The camera optical lens according to claim **16** further 10
satisfying the following conditions:

$$-1.85 \leq f_8/f \leq -0.86;$$

$$1.20 \leq (R15+R16)/(R15-R16) \leq 3.04; \text{ and}$$

$$0.05 \leq d15/TTL \leq 0.12.$$

18. The camera optical lens according to claim **1** further 15
satisfying the following condition: $TTL/IH \leq 2.91$;

where,

IH : an image height of the camera optical lens.

19. The camera optical lens according to claim **1**, wherein,
an FNO of the camera optical lens is less than or equal to 20
2.68;

where,

FNO: a ratio of an effective focal length of the camera
optical lens to an entrance pupil diameter.

* * * * *